Cuantificación de Incertidumbre en Clasificación Histopatológica

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Abstract

La adopción clínica de sistemas de aprendizaje profundo en anatomía patológica se ve limitada no tanto por la precisión de los modelos, sino por la incapacidad de estos de señalar cuándo no están seguros de sus predicciones. En este trabajo se comparan empíricamente tres modelos (dos de ellos con cuantificación de incertidumbre) que parten del mismo clasificador de referencia (ResNet18 preentrenada) aplicado al dataset *Histopathologic Cancer Detection*: (i) una red determinista entrenada como estimador máximo a posteriori (MAP); (ii) una aproximación de Laplace *post-hoc* sobre la última capa, con Hessiano factorizado por Kronecker (KFAC) y precisión del prior optimizada por verosimilitud marginal; y (iii) Monte Carlo Dropout con $T{=}50$ pasadas estocásticas en inferencia. Los tres modelos se evalúan en términos de discriminación (AUC-ROC, F1), calibración probabilística (Expected Calibration Error, Brier Score, log-verosimilitud negativa) y utilidad clínica mediante un análisis de triaje basado en incertidumbre epistémica.

1 Introducción

1.1 Motivación clínica y valor de negocio

La detección automática de células tumorales en parches de tejido histopatológico es uno de los casos de uso más consolidados del aprendizaje profundo en medicina. Los modelos convolucionales actuales alcanzan sistemáticamente un AUC superior a $0{,}95$ sobre benchmarks públicos, lo que ha motivado su integración progresiva en flujos de cribado asistido por ordenador. Sin embargo, este rendimiento oculta un problema estructural: un clasificador determinista emite una probabilidad puntual $\hat{p}(y{=}1 \mid \mathbf{x})$ sin ninguna cuantificación de la fiabilidad de esa estimación. Un caso morfológicamente ambiguo y un caso prototípico pueden recibir la misma probabilidad sin que el sistema distinga entre ellos.

Esta limitación es grave en dominios donde el coste de un falso negativo es desproporcionado y crítico, como puede ser la oncología. En este contexto, el enfoque bayesiano aporta una solución práctica: en lugar de aprender un único conjunto de pesos $\boldsymbol{\omega}^*$, el objetivo es inferir una distribución posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ sobre la que se marginaliza al predecir. La varianza de dicha distribución predictiva posibilita la toma de decisiones diferenciadas: cuando la incertidumbre supera un umbral preconfigurado τ_u, la muestra se deriva a revisión por un patólogo.

Esta regla operativa es la traducción directa del valor de negocio de estos modelos: no se trata solo de mejorar la precisión global, sino de identificar casos inciertos para que un experto (humano) los revise, reduciendo así el riesgo de falsos negativos sin sobrecargar al patólogo con casos típicos. En un despliegue real, las métricas online asociadas serían la tasa de derivación a revisión humana, la frecuencia con la que el patólogo rechaza la recomendación del modelo teniendo visibilidad de la incertidumbre (*user trust*), y la *reducción de falsos negativos* de la combinación modelo-patólogo frente a un patólogo sin apoyo algorítmico.

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1.2 Objetivos y contribuciones

Este trabajo persigue tres objetivos concretos. Primero, comparar cuantitativamente el rendimiento predictivo y la calibración de dos métodos de inferencia aproximada ampliamente utilizados en la literatura: la aproximación de Laplace (**?**) y Monte Carlo (MC) Dropout (**?**). En ambos casos, se parte de un mismo clasificador fijo (ResNet18 preentrenada) para aislar el efecto del método probabilístico. Segundo, analizar y aplicar la descomposición de la incertidumbre predictiva en sus componentes aleatoria y epistémica para definir una política de triaje. Tercero, estudiar los recursos computacionales de cada enfoque (tiempo de entrenamiento, latencia de inferencia, memoria) para contemplar un posible despliegue. El código, los pesos entrenados y una demostración interactiva en Streamlit están disponibles en este repositorio, reproducible con semilla aleatoria fija.

2 Formulación del problema

2.1 Modelo probabilístico

Disponemos de un conjunto de datos (dataset *Histopathologic Cancer Detection*) $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ con imágenes $\mathbf{x}_i \in \mathbb{R}^{3\times 96\times 96}$ y etiquetas binarias $y_i \in \{0, 1\}$ que indican la presencia o ausencia de tejido tumoral la región central (32x32) de la imagen. Modelizamos la verosimilitud como categórica sobre dos clases,

$$p(y \mid \mathbf{x}, \boldsymbol{\omega}) \;=\; \mathrm{Cat}\big(y \,\big|\, \mathrm{softmax}(f_{\boldsymbol{\omega}}(\mathbf{x}))\big), \tag{1}$$

donde $f_{\boldsymbol{\omega}} : \mathbb{R}^{3\times 96\times 96} \to \mathbb{R}^2$ es una red neuronal con pesos $\boldsymbol{\omega} \in \mathbb{R}^P$. Aunque el problema es binario, tomamos la parametrización *softmax* de dos dimensiones por cuestiones de compatibilidad con la librería `laplace-torch` (que asume salida categórica) y con el mecanismo de descomposición de la incertidumbre explicado en la SECCION4!.

Asumimos un prior gaussiano isotrópico sobre los pesos,

$$p(\boldsymbol{\omega}) \;=\; \mathcal{N}\big(\boldsymbol{\omega} \,\big|\, \mathbf{0}, \lambda^{-1}\mathbf{I}\big), \tag{2}$$

con precisión $\lambda > 0$. Para obtener la distribución predictiva bayesiana para una nueva observación $\mathbf{x}^\star$ necesitamos marginalizar sobre el posterior,

$$p(y^\star \mid \mathbf{x}^\star, \mathcal{D}) \;=\; \int p(y^\star \mid \mathbf{x}^\star, \boldsymbol{\omega})\, p(\boldsymbol{\omega} \mid \mathcal{D})\, \mathrm{d}\boldsymbol{\omega}. \tag{3}$$

Para ResNet18, con $P \sim 10^7$ parámetros, ni el posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ ni la integral (3) admiten forma cerrada. La metodología de este trabajo se centra en cómo aproximar esa integral manteniendo el mayor número posible de garantías bayesianas a un coste computacional aceptable.

2.2 Requisitos del sistema

Requisitos funcionales. El sistema debe (i) emitir, para cada entrada, una distribución predictiva $p(y \mid \mathbf{x})$ en lugar de una estimación puntual; (ii) descomponer la incertidumbre en sus componentes aleatoria (inherente al ruido de los datos, irreducible) y epistémica (asociada al conocimiento limitado del modelo, reducible con más datos); y (iii) producir derivar una muestra a revisión humana cuando la incertidumbre epistémica de su predicción supere el umbral preconfigurado τ_u.

Requisitos no funcionales. Establecemos como objetivos cuantitativos (i) un *Expected Calibration Error* (ECE) por debajo de 0.05; (ii) una latencia de inferencia por imagen que sea compatible con la posible integración en una interfaz de un programa médico; y (iii) un consumo de memoria que permita la ejecución en hardware de una clínica u hospital estándar. Estos requisitos condicionan, como veremos en la SECCION2! la elección del subconjunto de pesos sobre el que aplicar Laplace y la estructura adoptada para el Hessiano.

2.3 Selección de métodos

La literatura ofrece un rango amplio de métodos de inferencia aproximada para redes neuronales bayesianas (BNNs). La Tabla 1 resume los factores que más comprometen la selección, la cual debe ponderar el coste de entrenamiento, el coste de inferencia, la necesidad o no de reentrenamiento y la fidelidad de la aproximación al posterior.

Table 1: Compromisos entre métodos de inferencia aproximada en BNN. T denota el número de muestras en inferencia y M el tamaño de un *ensemble.*

Método	Coste entren.	Coste inf.	Reentren.	Cobertura del posterior
Laplace (*post-hoc*)	$1\times$ + 1 *fit*	T	No	Local, unimodal
MC Dropout	$1\times$	T	Sí	Local, unimodal
Inferencia variacional (MFVI)	$\sim 2\times$	T	Sí	Unimodal
Deep Ensembles	$M\times$	M	Sí	Multimodal
MCMC (SGLD, HMC)	$\gg 1\times$	T	Sí	Asintóticamente exacta

Tras analizar los modelos estudiados, se optó por seleccionar la aproximación de Laplace y el enfoque de MC Dropout para compararlos en este estudio. El primero es un método *post-hoc* que aprovecha la curvatura real del *loss* en el punto MAP, y el segundo es trivial de implementar sobre cualquier arquitectura que ya incluya *dropout*. Ambos comparten la misma arquitectura troncal y, en el caso de Laplace, el mismo punto MAP, lo que permite atribuir las diferencias observadas casi exclusivamente al mecanismo de cuantificación de incertidumbre. Deep Ensembles y MCMC quedan fuera del alcance por coste computacional; y MFVI se descarta porque requeriría una reimplementación completa del entrenamiento sin asegurar una ventaja teórica sobre Laplace en términos de fidelidad del posterior.

3 Antecedentes

3.1 Del estimador MAP al posterior

El entrenamiento estándar de una red neuronal con pérdida *cross-entropy* y regularización L2 admite una interpretación bayesiana exacta. La función de pérdida que minimiza nuestro `Trainer` es

$$\mathcal{L}(\boldsymbol{\omega}) \;=\; -\sum_{i=1}^{N} \log p(y_i \mid \mathbf{x}_i, \boldsymbol{\omega}) \;+\; \tfrac{\lambda}{2}\|\boldsymbol{\omega}\|_2^2, \tag{4}$$

donde el segundo término coincide, salvo una constante , con $-\log p(\boldsymbol{\omega})$ bajo el prior (2). Por ende, el mínimo de (4) coincide con el estimador MAP,

$$\boldsymbol{\omega}^* \;=\; \arg\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \;=\; \arg\max_{\boldsymbol{\omega}} \big[\log p(\mathcal{D} \mid \boldsymbol{\omega}) + \log p(\boldsymbol{\omega})\big]. \tag{5}$$

En concreto, el hiperparametro λ controla la precisión del prior y, por tanto, el grado de regularización. La equivalencia mostrada en (5) es clave: demuestra que el MAP obtenido por Adam con *weight decay* no es un punto de partida arbitrario, sino que es el *mode* (pico) del posterior bajo el priori adoptado, y por tanto, el punto alrededor del cual Laplace construye su aproximación gaussiana.

3.2 Aproximación local vs. aproximación variacional

Las dos estrategias implementadas responden a filosofías distintas para aproximar $p(\boldsymbol{\omega} \mid \mathcal{D})$. La aproximación de Laplace es una técnica de optimización local: fija una gaussiana centrada en $\boldsymbol{\omega}^*$ y ajusta su matriz de covarianza a la curvatura (Hessiano) del log-posterior en ese punto. Por el contrario, MC Dropout es una técnica particular de inferencia variacional cuya familia $q(\boldsymbol{\omega})$ es de tipo mezcla de Bernoulli **?**. En este caso, el entrenamiento estándar con *dropout* y *weight-decay* ya maximiza el ELBO asociado a esa familia, y las pasadas estocásticas en inferencia corresponden a muestras del posterior aproximado $q(\boldsymbol{\omega})$.

Ambos enfoques producen un procedimiento para muestrear pesos y aproximar la integral (3) mediante Monte Carlo,

$$p(y^\star \mid \mathbf{x}^\star, \mathcal{D}) \;\approx\; \frac{1}{T}\sum_{t=1}^{T} p(y^\star \mid \mathbf{x}^\star, \boldsymbol{\omega}_t), \tag{6}$$

La diferencia reside en la definición de $p(\boldsymbol{\omega})$ y en el procedimiento para obtener las muestras $\boldsymbol{\omega}_t$, tal y como se explicará en la SECCION 4!.

4 Configuración experimental

5 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

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Please pay special attention to the instructions in Section 7 regarding figures, tables, acknowledgments, and references.

6 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

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6.1.1 Headings: third level

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The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated…

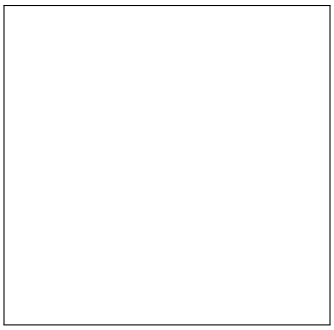

Figure 1: Sample figure caption.

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2025` package:

```
\PassOptionsToPackage{options}{natbib}
```

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```
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```

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Note that footnotes are properly typeset *after* punctuation marks.[2]

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Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

[1] Sample of the first footnote.

[2] As in this example.

Table 2: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 2.

7.5 Math

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- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

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8.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

Acknowledgments and Disclosure of Funding

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[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above), or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.

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  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
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